MILITARY SPECIFICATION

FUZE, GUIDED MISSILE, FMU-141/B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the FMU-141/B Guided Missile Fuze, referred to herein as the fuze.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81	Tag, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1336

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L-T-100	Tape, Pressure-Sensitive Adhesive, Polyester Film.
TT-I-735	Isopropyl Alcohol.

MILITARY

MIL-S-6872	Soldering Process, General Specification for.
MIL-T-14757	Trichlorotrifluoroethane.
MIL-T-18303	Test Procedures, Reproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.
MIL-C-21723	Composition CH-6.
MIL-D-85294	Detonator, Electric, MARK 71 MOD 1.
MS51938	Seal, Metallic, Lead and Wire.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1246	Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1316	Fuze, Design Safety Criteria for.

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MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications of the issue specified in the solicitation, form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Naval Air Systems Command
(Code Ident 30003)

WS-2183	Critical Item Product Fabrication Specification for Triggering Device, Fuze, MARK 4 MOD 0.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

Naval Sea Systems Command
(Code Ident 53711)

WS-1639	Detonator, Electric, MARK 85 MOD 0.
OS-8164	Descriptions and Requirements for Lead, Explosive, MARK 9 MOD 0.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

642AS5610	Fuze, Guided Missile, FMU-141/B.
642AS5639	Shipping and Storage Container, Packing and Marking.
SA497752	Test Set, Fuze and Probe (Harpoon).

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SA497753	Fixture, Explosive Train Drop.
SA498016	Fixture, Vibration.
SA498017	Fixture, Jolt.
SA498018	Fixture, Forty-Foot Drop.
SA498019	Fixture, Transportation Vibration.
2049577	Cup, Booster.
SA2876062	Fixture, Shock.

PUBLICATIONS

Naval Weapons Center
(Code Ident 12934)

Reg Memo 335-196-76	Harpoon Guided Missile Fuze and Pressure Probe Test Set SA497752 Operating Instructions.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 108	Steel Bars, Carbon, Cold-Finished, Standard Quality.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS.

3.1 Item description. The fuze was designed in accordance with MIL-STD-1316 and is an electromechanical device that maintains an out-of-line explosive train and an electrical firing circuit in the safe (open) position until certain prerequisite events have occurred after missile launch. The general configuration of the fuze is shown on Figure 1.

3.1.1 Government-furnished property.

3.1.1.1 Autonomous components. When specified in the contract or purchase order, the Government will furnish the contractor with the following components for each fuze produced under a specified contract (see 6.2.1):

a. Four MARK 9 MOD 0 Explosive Leads (see OS-8164).

b. One MARK 4 MOD 0 Triggering Device (see WS-2183).

c. Two MARK 85 MOD 0 Electric Detonators (see WS-1639).

d. Two MARK 71 MOD 1 Electric Detonators (see MIL-D-85294).

3.1.1.2 Support equipment. When specified in the contract or purchase order, the Government will furnish the independent test facility designated in the contract or purchase order with the following support equipment to perform the environmental tests specified herein (see 6.2.1):

a. One explosive train drop fixture (see Drawing SA497753).

b. One vibration fixture (see Drawing SA498016).

c. One jolt fixture (see Drawing SA498017).

d. One forty-foot drop fixture (see Drawing SA498018).

e. One transportation vibration fixture (see Drawing SA498019).

f. One shock fixture (see Drawing SA2876062).

3.1.2 Government-loaned property. When specified in the contract or purchase order, the Government will loan the contractor and the independent test facility designated in the contract or purchase order one Fuze and Probe Test Set, Harpoon (see SA497752), referred to herein as the test set (see 6.2.1).

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4. and 6.2.1).

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3.3 Design and construction.

3.3.1 Production drawings. The fuze shall meet the requirements of and be manufactured and assembled in accordance with the drawings and documents listed on Drawing 642AS5610 (see 4.7.9).

3.3.2 Standards of manufacture.

3.3.2.1 Product cleanliness. Product cleanliness shall meet the requirements of MIL-STD-1246, Level 200. No oils or solid lubricants of any kind shall be used on any part of the fuze after final cleaning of components, except as specified on the drawings or herein.

3.3.2.2 Working environment. The contractor shall provide facilities for the fabrication, assembly, and testing of the fuze to be delivered in accordance with this specification. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

3.3.2.3 Soldering. Procedures and requirement for preparation and soldering of electrical connections shall be in accordance with WS-6536. Mechanical soldering shall be in accordance with MIL-S-6872.

3.4 Performance characteristics.

3.4.1 Electrical. The requirements of 3.4.1.1 and 3.4.1.2 shall be met by the fuze during and after assembly in the configurations shown in Table I. In Tables II, III, IV, and V, switch position numbers refer to the test set (see 3.1.2), and correspond to the pin numbers in the fuze connector. An electrical schematic for the fuze is shown on Figure 2.

TABLE I. Assembly configurations for electrical requirements.

Configuration	Detonators and R16, 17, 18 installed	Welded[1/]	Sealed[1/]
A	No	No	No
B	Yes	Yes	No
C	Yes	Yes	Yes

[1/] See Drawing 642AS5610.

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TABLE II. Resistance (safe condition) requirements for configuration A.

Test	Ohmmeter scale[1/] reading	Test set switch positions		Resistance[1/] During manufacture		Items checked (see Figure 2)
		C+	C-	Lower limit	Upper limit	
S1	10M	12	8	Open	...	S1A
S2	10M	9	8	Open	...	S1B
S3	1M	5	8	270K	390K	R2, 3, 5, D4, S1C
S4	1M	8	5	435K	645K	R2, 3, S1C, D3
S5	1M	4	8	175K	725K	R4, S1C
S6	1M	8	4	850K	1.25M	R4, S1C
S7	100K	7	8	22K	52K	D1, S1C
S8	100K	10	8	23K	39K	R1, 12, 13, 14 D2, S1C
S9[2/]	100K	10	2	23K	32K	R12,14, S1E
S10	10K	2	8	4.6K	6.1K	R13
S11	1K	6	11	295 ohms	365 ohms	R15
S12	100 ohms	11	6	60 ohms	111 ohms	R15, Q5C
S13	1 ohm	10	6	...	...	...
S13 (cal)[3/]	1 ohm	10	6	...	...	...
S13 (net)[4/]	...	...	...	0 ohm	1 ohm	S1D

[1/] Readings are obtained by using a Hewlett-Packard VOM Model 3469B (part of test set SA497752).

[2/] Depress UMBILICAL SHORT switch (disconnects pin 11 from ground) (part of test set SA497752).

[3/] Calibration readings are taken at calibration socket (part of test set SA497752).

[4/] Net values are the standard readings minus the calibration readings.

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TABLE III. Resistance (safe condition) requirements for configurations B and C.

Test	Ohmmeter scale[1/] reading	Test set switch positions		Resistance[1/]				Items checked (see Figure 2)
				Configuration				
				During manufacture B and C		After shipment, C		
		C+	C-	Lower limit	Upper limit	Lower limit	Upper limit	
S1	100K	12	8	15K	19K	14K	20K	S1A
S2	100K	9	8	15K	19K	14K	20K	S1B
S3	1M	8	5	435K	645K	420K	660K	R2, 3, S1C, D3
S4	1M	8	4	850K	1.25M	800K	1.30K	R4, S1C
S5	1K	6	11	295 ohms	365 ohms	260 ohms	400 ohms	R15
S6	100 ohms	11	6	60 ohms	111 ohms	55 ohms	116 ohms	R15, Q5C
S7	1 ohm	10	6	...	...	...	...	...
S7 (cal)[3/]	1 ohm	10	6	...	...	...	...	...
S7 (net)[4/]	...	...	...	0 ohm	1 ohm	0 ohm	1.2 ohms	S1D

[1/] Readings are obtained by using a Hewlett-Packard VOM Model 3469B (part of test set SA497752).

[2/] Depress UMBILICAL SHORT switch (disconnects pin 11 from ground) (part of test set SA497752).

[3/] Calibration readings are taken at calibration socket (part of test set SA497752).

[4/] Net values are the standard readings minus the calibration readings.

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TABLE IV. Resistance (armed condition) requirements for configuration A.

Test	Ohmmeter scale reading 1/	Test set switch positions		Resistance 1/		Item checked (see Figure 2)
		C+	C-	Lower limit	Upper limit	
A1	1M	12	8	Open	...	S1A
A2	1M	9	8	Open	...	S1B
A3	1M	7	8	600K	1.4M	R2, 4, 5, 11, D1, 4, Q2
A4	1M	10	8	140K	185K	R1, 4, 11, 13, 14, Q2, D2
A5	1M	2	10	142K	170K	R14, S1E

1/ Readings are obtained by using a Hewlett-Packard VOM Model 3469B (part of test set SA497752).

3.4.1.1 Resistance (safe condition). While in the safe condition (see 6.3.1), the fuze shall meet the resistance requirements of Table II for configuration A, Table I, and the requirements of Table III for configurations B and C, Table I (see 4.7.1.1).

3.4.1.2 Resistance (armed condition). While in the armed condition (see 6.3.2), the fuze shall meet the resistance requirements of Table IV for configuration A, Table I, and the requirements of Table V for configurations B and C, Table I (see 4.7.1.2).

3.4.2 Mechanical operability. The fuze shall meet the mechanical operability requirements of Table VI for each of the three configurations (A, B, and C) specified in Table I. Beginning and ending times are indicated by the beginning and ending switching functions (see 4.7.2). (The requirements of Table VI represent fuze calibration times and not fuze arming times.)

3.4.3 Seal. At the contractor's facility, the leakage rate of the fuze shall be not greater than 5×10^{-5} standard atmosphere cubic centimeters per second (std atm cm^3/s) when measured within 4 hours after pressurizing and sealing. Anytime after shipment from the contractor's facility, the leakage rate of the fuze shall be not greater than 5×10^{-4} std atm cm^3/s (see 4.7.3).

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TABLE V. Resistance (armed condition) requirements for configuration B.

Test	Ohmmeter scale reading 1/	Test set switch positions		Resistance 1/				Items checked (see Figure 2)
				Configuration				
				B		C 4		
		C+	C-	Lower limit	Upper limit	Lower limit	Upper limit	
A1	10 ohms	12	8	...	...	...	...	...
A1 (cal) 2/	10 ohms	12	8	...	...	...	...	...
A1 (net) 3/	...	...	...	0 ohm	5 ohms	0 ohm	6 ohms	SQ1, SQ2, S1A
A2	10 ohms	9	8	...	...	...	...	...
A2 (cal) 2/	10 ohms	9	8	...	...	...	...	...
A2 (net) 3/	...	...	...	0 ohm	5 ohms	0 ohm	6 ohms	SQ3, SQ4, S1B
A3	100K	7	8	14K	21K	13K	22K	S1C
A4	100K	10	8	12K	19K	11K	20K	R1, 4, 11, 13, 14, D2, Q2
A5	1M	2	10	130K	185K	125K	190K	R14, S1E

1/ Readings are obtained by using a Hewlett-Packard VOM Model 3469B (part of test set SA497752).

2/ Calibration readings are taken at calibration socket (part of test set SA497752).

3/ Net values are the standard readings minus the calibration readings.

4/ These resistance checks performed during environmental test only.

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TABLE VI. Mechanical operability requirements for configurations A, B, and C.

Test	Pressure applied	Solenoid Vdc		Requirements	
		Configuration			
		A and B (Vdc)	C (Vdc)	A and B	C
All go	33.0 ±0.5 inches water	22 ±1	....	Arm[1/]	Arm[1/]
All go	40.0 ±0.5 inches water	...	22 ±1	Arm[1/]	Arm[1/]
No go	25.0 ±0.5 inches water	24 ±1	28 ±1	Remain safe	Remain safe
Timing	2.0 ±0.1 lb/in^2	24 ±1	28 ±1	11.0 ±1.5 seconds[2/]	11.75 ±2.25 seconds[2/]
Burst	10.0 ±0.3 lb/in^2	24 ±1	28 ±1	Arm[1/]	Arm[1/]

[1/] Rotor fully rotated and locked.

[2/] Rotor fully rotated and locked with time measured as shown on Figure 3.

3.4.4 Safety determination. The fuze shall be determined to be in the safe condition (see 6.3.1) if a capacitor (5.6 microfarad (μF) minimum), charged to 28 ±1 volts direct current (Vdc) is connected across pins 12(+) and 8(-) of the connector and no explosive component initiates (see 4.7.4).

3.4.5 Environmental conditions. The fuze shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the environments specified in 3.4.5.1 through 3.4.5.5.

3.4.5.1 Transportation vibration-temperature. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.7.5.1.

3.4.5.2 Shock.

3.4.5.2.1 Handling shock. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.7.5.2.1.

3.4.5.2.2 Pyro shock. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.7.5.2.2.

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3.4.5.2.3 Thermal shock. The fuze shall remain safe (see 6.3.3) during the test and be operable (see 6.3.4) after exposure to the environment of 4.7.5.2.3.

3.4.5.3 Temperature and humidity. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.7.5.3.

3.4.5.4 Captive flight and boost flight vibration (safe condition). The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment of 4.7.5.4.

3.4.5.5 Missile flight vibration (armed condition). The fuze shall remain armed, explosive components shall not initiate, and the switches or associated components shall not fail in any way as to interfere with initiation of the leads during and after exposure to the environment of 4.7.5.5.

3.4.6 Environmental safety.

3.4.6.1 Jolt. The fuze shall be safe to handle, store, and transport after exposure to the environment of 4.7.6.1. During exposure, no explosive component shall initiate. After exposure, an attempt shall be made to arm the fuze in accordance with 4.7.2. The fuze can fail to arm or, if arming occurs the fuze arming time shall be not less than the minimum time specified in Table VI.

3.4.6.2 Twelve meter drop. The fuze shall meet the requirements specified below after exposure to the environment of 4.7.6.2:

a. No explosive components beyond the detonator shall have initiated.

b. No loose explosive powder shall be present in the housing (see Drawing 642AS5551).

c. The lead rotor (see Drawing 642AS5422) shall be in the safe position (see 6.3.1) and unable to be turned to the armed position manually (see 6.3.2).

d. The piston shaft (see Drawing 642AS5495) shall be in the safe position (see 6.3.1) and locked by the piston lock (see Drawing 642AS5538) and the solenoid cam (see Drawing 642AS5334-2).

e. The fuze shall be safe to handle and dispose of.

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3.4.7 Explosive train.

3.4.7.1 Static detonator safety. There shall be no evidence of detonation, burning, charring, scorching, melting, or rupturing of the explosive leads when the detonators are initiated with the arming rotor in the safe position (see 6.3.1). Smudging of the surfaces of the leads, due to deposits resulting from initiation of the detonators, is acceptable. The fuze housing shall not rupture, and any particles of explosive output ejected from the fuze housing through the viewing window shall not be hazardous to personnel or equipment (see 4.7.7.1)

3.4.7.2 Explosive-train output. The explosive train of the fuze, in the instantaneous mode (see 6.3.5), the delay mode (see 6.3.7), and the alternate mode (see 6.3.8), shall produce an average depth of dent of 0.030 inch minimum when armed and detonated on a steel dent block (see Figure 4 and 4.7.7.2).

3.5 Visual. The fuze shall meet the following requirements (see 4.7.8):

a. There shall be no evidence of external damage, such as cracks, broken windows, dents, or other blemishes, which may interfere with functioning or fit at the next assembly.

b. The rotor shall be in the safe position (see 6.3.1), as evidenced by a white letter "S" on a green background centered in the window located on the fuze housing.

c. The safety inspection tag (see Drawing 642AS5603) shall be properly signed by both the contractor and Government inspectors, and securely attached to the fuze housing by a lead and wire seal in accordance with MS51938-5, Type I, to indiçate that a successful safety test (see 3.4.4) has been performed. The contractor's identification symbol shall be pressed into the lead seal.

3.6 Workmanship. The fuze and all components shall meet the workmanship requirements of MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform

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or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The contractor shall ensure that test equipment and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.1.1.1 Test equipment calibration. All test equipment shall be calibrated and maintained in accordance with the requirements of MIL-STD-45662. When specified in the contract or purchase order, the contractor shall prepare a calibration plan in accordance with the Data Item Description (DID) cited in 6.2.2. The calibration plan shall be acceptable to the Government prior to use.

4.1.1.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used in the performance of the specified tests shall have an accuracy of at least 10 percent of the tolerance for the variables to be measured.

4.1.1.3 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein or supplied by the Government (see 3.1.1), are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.1.1.1 and 4.1.1.2 and are subject to acceptance by the procuring activity.

4.1.1.4 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. Unless otherwise specified in the contract or purchase order, the procuring activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.4).

b. Quality conformance inspections (see 4.5).

c. Individual unit acceptance inspections (see 4.6).

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4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 3.3.2.2.

4.3.1 Resetting. Before detachment from the test set, all fuzes shall be reset to the safe position (see 6.3.1) after any test where arming occurs, except for those fuzes which must be left in the armed position (see 6.3.2) for subsequent tests. Although failure of a fuze to reset to the safe position does not affect acceptance of a lot, any fuze which cannot be reset shall be rejected. Unless otherwise specified in the contract or purchase order, disposition of armed fuzes that repeatedly fail to reset to the safe position shall be at the discretion of the cognizant demolition officer (see 6.2.1). The unsafe condition shall be indicated by securely attaching an unsafe condition tag as shown on Figure 5.

4.3.2 Test plan and test procedures. When specified in the contract or purchase order, the contractor shall prepare, using MIL-T-18303 as a guide, a test plan and test procedure in accordance with the DID cited in 6.2.2. The test plan and test procedure shall be acceptable to the Government prior to use.

4.3.3 Lot inspection report. When specified in the contract or purchase order, the contractor shall prepare a lot inspection report for each lot produced under a specified contract, including first article, inspection, and rejected lots, in accordance with the DID cited in 6.2.2. This report shall contain the data necessary for the technical monitoring of lot inspection.

4.3.4 Production summary report. When specified in the contract or purchase order, the contractor shall prepare a production summary report in accordance with the DID cited in 6.2.2. The report shall contain the data necessary for monitoring production by the Government.

4.3.5 Fuze inspection record. When specified in the contract or purchase order, an inspection record, as shown on Figure 6 and prepared in accordance with the DID cited in 6.2.2, shall be properly filled in for each fuze produced under a specified contract. The required test data shall be extracted from the inspection and acceptance records maintained by the contractor.

4.3.6 Inspection and acceptance records. Inspection and acceptance records shall be maintained by the contractor and shall be organized to facilitate search and review. These records shall include the results of all tests and inspections specified herein and any additional tests, inspections, repairs, rework, replacements, or retests of any part, subassembly, assembly, or end-item deemed necessary by the contractor to achieve acceptance. These records shall be made available to the procuring activity on request.

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4.4. First article inspections. When specified in the contract or purchase order (see 6.2.1), a first article sample of 30 fuzes shall be subjected to the inspection sequence of 4.4.1. The first article sample shall be manufactured using the same methods, materials, equipment and processes proposed for production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts. Any production prior to approval of the first article shall be at the risk of the contractor.

4.4.1 First article inspection sequence. The first article sample shall be subjected to the individual unit acceptance inspections of Figure 7 (see 4.6). The first article sample shall then be subjected to the inspections of Figures 8 and 9 in the sequence shown. The inspections of Figure 9 shall be performed at an independent facility designated in the contract or purchase order (see 6.2.1).

4.4.2 First article acceptance and rejection criteria. When subjected to the inspection sequences of Figures 8 and 9, each fuze of the first article sample shall meet the applicable performance requirements of Section 3 on the first and only try. Defects are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified herein and on the drawings. The following shall be cause for refusal to grant first article approval:

a. One or more CRITICAL defects are found, or

b. Two or more MAJOR defects are found.

4.5 Quality conformance inspections. The tests and examinations of 4.7 shall be performed to determine that all the requirements of Section 3 have been met. Tests shall be performed in accordance with the test plan and test procedures which have been approved by the Government (see 4.3.2).

NOTE: The contractor may, prior to forming and submitting a lot for acceptance, subject the completely assembled fuze to the inspections of 4.7.1 through 4.7.4, 4.7.8, and 4.7.9 to the extent deemed necessary.

4.5.1 Inspection lot. Unless otherwise specified in the contract or purchase order, an inspection lot shall contain not less than 51 nor more than 280 fuzes, including those required for destructive tests (see 6.2.1). An inspection lot, as defined in MIL-STD-105, shall consist of completed fuzes fabricated under the same conditions and offered for inspection at the same time.

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4.5.1.1 Serial numbers. Unless otherwise specified in the contract or purchase order, all fuzes shall be assigned a Government serial number in order of consecutive assembly (see 6.2.1). After assignment of a Government serial number to each fuze, the inspection lot shall be formed and identified by lot number.

4.5.2 Quality conformance inspection sequence. Prior to formation of the lot, the fuze shall be subjected to the individual unit acceptance inspections of Figure 7 (see 4.6). After assignment of serial numbers, and after the inspection lot has been formed and identified by number, the lot shall be subjected to the inspections of Figure 10. After completion of the inspections of Figure 10, 15 fuzes shall be randomly sampled from the inspection lot in accordance with MIL-STD-105 and subjected to the inspections of Figure 9 at a facility designated in the contract or purchase order (see 6.2.1). The remainder of the lot shall be held in bond pending acceptance of the 15-fuze sample.

4.5.3 Quality conformance acceptance and rejection criteria. Defects in the fuze are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified herein and on the drawings. Acceptance and rejection shall be based on the following:

a. For the inspections of Figure 7, all parts, subassemblies and assemblies found to be defective shall be rejected.

b. For the inspections of Figure 10:

(1) Where 100 percent inspection is required, all defective fuzes shall be rejected.

(2) Where sampling is required for lot sizes of 51 to 159 fuzes, lot acceptability shall be determined in accordance with MIL-STD-105, Single Sampling Plan, Tightened Inspection, sample size code letter H, acceptable quality level (AQL) of 0.40 percent defective. For lot sizes of 160 to 280 fuzes, lot acceptability shall be determined in accordance with MIL-STD-105, Double Sampling Plan, Normal Inspection, sample size code letter K, AQL of 0.40 percent defective.

c. For the inspections of Figure 9, the failure of any sample fuze to meet the applicable requirements of Section 3 shall cause rejection of the lot.

4.5.3.1 Counting of defects. When counting defects found in fuzes to determine acceptability of the lot, fuzes with multiple defects shall be

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counted as a single defect, using the highest classification of the defects found. For example, when fuze has both a CRITICAL and MAJOR defect, it shall be counted as a CRITICAL defect.

4.6 Individual unit acceptance inspections. Individual unit acceptance inspections shall be performed during the manufacture of the fuze in accordance with Figure 7. Parts, subassemblies, or assemblies found to be defective during the inspection sequence shall be rejected.

4.7 Methods of inspection.

WARNING

Unless otherwise specified, verify that the fuze is in the safe condition (see 6.3.1) before removing the fuze from the test set at any time during any test sequence. Discharge the firing capacitor by shorting pins 5 and 8 together for 2 minutes minimum. Verify that the firing capacitor has been discharged by measuring the voltage across pins 5 and 8; if less than 1 Vdc, the fuze is safe to handle.

4.7.1 Electrical. The tests of 4.7.1.1 and 4.7.1.2 shall be conducted on fuzes during and after assembly as specified on the applicable figures. Resistance measurements shall be performed using the test set (see 3.1.2 and 3.4.1), or standard test equipment that meets the requirements of 4.1.1. Required test data shall be recorded on the inspection record (see Figure 6).

4.7.1.1 Resistance (safe condition). To determine conformance to 3.4.1.1, verify the resistance requirements of Tables II and III with the fuze in the safe condition (see 6.3.1). The resistance shall be within the limits specified in the table.

M101

4.7.1.2 Resistance (armed condition). To determine conformance to 3.4.1.2, verify the resistance requirements of Tables IV and V with the fuze in the armed condition (see 6.3.2). The tests for configuration C (see Table I) of Table V shall be performed during the environmental inspection sequence of Figure 9. The resistance shall be within the limits specified in the tables.

M102

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4.7.2 Mechanical operability. To determine conformance to 3.4.2, verify the mechanical operability requirements of Table VI using the test set described in Reg Memo 335-196-76 (see 3.1.2 and 3.4.1), or standard test equipment with a two-channel (minimum) recorder and acoustic pickup, which meets the requirements of 4.1.1. Timing shall be determined from the output record chart as shown on Figure 3. Record test results on the inspection record (see Figure 6).

NOTE: An acoustic pickup shall be used as an audio aid to verify fuze arming and resetting (see 4.3.1).

Timing failure on min side:	C1
Timing failure on max side:	M103
Operability otherwise:	M104

4.7.3 Seal. To determine conformance to 3.4.3, test the fuze in accordance with MIL-STD-202, Method 112, Test Condition C, Procedure IV. Trichlorotrifluoroethane (MIL-T-14757) or isopropyl alcohol (TT-I-735, grade optional) may be used as alternative immersion fluids for gross leak testing. At the contractor's facility, the fuze shall be tested within 4 hours after pressurizing and sealing and the results shall be recorded on the inspection record (see Figure 6).

M105

4.7.4 Safety determination. To determine conformance to 3.4.4, test the fuze as follows:

a. If required, reset the fuze to the safe condition (see 6.3.1) using the test set (see 4.3.1). Discharge the firing capacitor by shorting pins 5 and 8 together for 2 minutes minimum. Verify that the firing capacitor has been discharged by measuring the voltage across pins 5 and 8; if less than 1 Vdc, the fuze is safe to handle.

b. Visually verify that the fuze is in the safe condition (see 6.3.1), as evidenced by a white letter "S" on a green background centered in the window located on the fuze housing.

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WARNING

Do not proceed with the test unless the fuze is in the safe condition (see 6.3.1). The following test will cause initiation of the explosive train if the fuze is not in the safe condition.

c. Attach a firing cable to pins 12(+) and 8(-) of the connector.

d. Place the fuze in a safety chamber.

e. Connect a firing cable to a capacitor discharge system capable of delivering a minimum of 20,000 ergs to the fuze (28 ±1 Vdc and 5.6 μF).

f. Discharge the charged capacitor into the firing cable.

g. Run a continuity check between pins 12(+) and 8(-) to determine if the fuze is safe to handle (open indication on the 1-kilohm (kohm) scale of the test set).

h. Remove the fuze from the safety chamber. Visually verify that the fuze is in the safe condition (see 6.3.1) as specified in b above.

i. Fill in, sign, and securely attach a safety inspection tag (see Drawing 642AS5603) to the housing using a lead and wire seal (see 3.5c).

j. Record the test results on the inspection record (see Figure 6).

C2

4.7.5 Environmental. All fuzes of the first article sample, or a randomly selected sample of 15 fuzes from an inspection lot, shall be subjected to the environmental tests specified in 4.7.5.1 through 4.7.5.5 in accordance with Figure 9 at an independent test facility designated in the contract or purchase order (see 6.2.1). Any fuze subjected to these tests and the tests of 4.7.6 and 4.7.7 shall be considered destroyed and shall not be returned to the lot nor forwarded for service use, but shall be labeled appropriately and held for disposal as specified in the contract or purchase order (see 6.2.1). During these tests and the tests of 4.7.6 and 4.7.7, initiation of any explosive component, except those which are initiated intentionally as part of a test, shall be considered a CRITICAL defect and shall cause rejection of the lot from which the fuze was drawn. Throughout the tests, the fuze shall be in the safe condition (see 6.3.1) unless otherwise specified herein or on the drawing.

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CAUTION

The explosive components of the fuze are highly sensitive and hazardous; proper safety precautions must be observed during handling (see 6.4).

4.7.5.1 Transportation vibration-temperature. To determine conformance to 3.4.5.1, mount the fuze to the vibration fixture (see Drawing SA498019). Test the fuze in accordance with MIL-STD-331, Test 119, Procedure 2.

Remain safe: C3 Remain operable: M106

NOTE: For the vibration and shock tests specified herein, the test fixtures shall be mounted to a vibration table or shock machine and the mounting bolts shall be tightened to a torque of 25 ±1 foot-pounds. The fuze ring nut shall be tightened to a torque of 100 ±20 foot-pounds.

4.7.5.2 Shock.

4.7.5.2.1 Handling shock. To determine conformance to 3.4.5.2.1, mount the fuze to the shock fixture (see Drawing SA2876062; alternate, vibration fixture of Drawing SA498016). Test the fuze in accordance with MIL-STD-202, Method 213, Test Condition J, except that the amplitude shall be 15 ±1 gravity units (g).

Remain safe: C4 Remain operable: M107

4.7.5.2.2 Pyro shock. To determine conformance to 3.4.5.2.2, test the fuze in accordance with the following. Subject the fuze to one transient oscillation in the primary axis (see Figure 1) with the shock response spectrum shown in Figure 11, and one transient oscillation in each of the two remaining orthogonal secondary axes with the shock response spectrum shown in Figure 11. The duration of the applied shock transients shall be between 0.05 and 0.5 second. The analyses used to determine the shock response spectrum shall be performed at a sequence of natural frequencies at intervals of one sixth octave or smaller. The allowed tolerance will be ±6 dB from the specified spectrum except for any given range of frequencies equal to one octave, the test levels shall be within ±3 dB of the specified frequency range for at least one half of the octave.

Remain safe: C5 Remain operable: M108

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4.7.5.2.3 Thermal shock. To determine conformance to 3.4.5.2.3, test the fuze in accordance with MIL-STD-810, Method 503.1, 2 cycles.

Remain safe: C6 Remain operable: M109

4.7.5.3 Temperature and humidity. To determine conformance to 3.4.5.3, test the fuze in accordance with MIL-STD-331, Test 105, either method. The connector shall be thoroughly cleaned and dried after the test.

Remain safe: C7 Remain operable: M110

4.7.5.4 Captive flight and boost vibration (safe condition). To determine conformance to 3.4.5.4, mount the fuze to the vibration fixture (see Drawing SA498016). Test the fuze in accordance with MIL-STD-810, Method 514, Procedure V, Part 2. For the captive flight vibration portion, use curve shown on Figure 12 herein, except random vibration shall be applied for 2 hours minimum along each of the three mutually perpendicular axes for a total of 6 hours. Boost flight vibration shall be in accordance with Figure 13. Random vibration shall be applied for 1 minute minimum along each of three mutually perpendicular axes for a total of 3 minutes.

Remain safe: C8 Remain operable: M111

4.7.5.5 Missile flight vibration (armed condition). To determine conformance to 3.4.5.5, mount the armed fuze to the vibration fixture (see Drawing SA498016). Test the fuze in accordance with MIL-STD-810, Method 514, Procedure V, Part 2, Curve shown on Figure 12 herein. Random vibration shall be applied for 10 minutes minimum along each of the three mutually perpendicular axes for a total of 30 minutes. Throughout the test, the fuze shall be electrically monitored to determine whether it remains in the armed condition (see 6.3.2).

Remain armed and operable: M112

4.7.6 Environmental safety. The conditions of 4.7.5 shall apply for the tests of 4.7.6.1 and 4.7.6.2.

4.7.6.1 Jolt. To determine conformance to 3.4.6.1, mount the fuze to the jolt fixture (see Drawing SA498017). Test the fuze in accordance with MIL-STD-331, Test 101.2, Procedure II. After jolt, test the fuze in accordance with 4.7.2.

Remain safe and minimum arming time: C9

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4.7.6.2 Twelve meter drop. To determine conformance to 3.4.6.2, mount the fuze to the forty-foot drop fixture (see Drawing SA498018). Test the fuze in accordance with MIL-STD-331, Test 103.2, Procedure 1. When only three fuzes are to be dropped (see Figure 9), the drop position shall be vertical (connector up), horizontal, and 45 degrees from vertical (connector down).

Remain safe and be safe to dispose of: C10

4.7.7 Explosive train. The conditions of 4.7.5 shall apply for the tests of 4.7.7.1 and 4.7.7.2.

4.7.7.1 Static detonator safety. To determine conformance to 3.4.7.1, test the fuze in accordance with MIL-STD-331, Test 115, and as follows:

WARNING

Proper safety precautions must be observed when making the following modifications and performing the following test (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials.

a. Verify that the fuze meets the safe condition requirements of 4.7.4a and b.

b. For explosive output witness, wrap the fuze with one layer of polyester pressure-sensitive tape (L-T-100, Type I), centered over the viewing window in the housing.

c. Attach a shorting plug (all pins shorted) to the connector.

d. Machine off the welded seal around the flange of the fuze and remove the internal assembly from the housing. Place the assembly behind a protective shield.

e. Modify terminal board TB1 in the following sequence and as shown on Figure 14.

(1) Remove diode D5 from terminal board TB1 between terminals 3 and 4.

(2) Remove thermofit insulation from terminal board TB1, terminals 3 and 4. Remove solenoid wires and clean terminals.

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(3) Add 22-gauge jumper wire from terminal 1 to terminal 6.

(4) Add 22-gauge jumper wire from terminal 1 to terminal 3.

f. Reinsert the internal assembly into the housing. Clamp the internal assembly and the housing together in such a manner to prevent separation and movement during the test.

g. Place the fuze into a firing chamber capable of containing 2.2 grams of CH-6 (see MIL-C-21723) explosive output.

h. Remove the shorting connector and connect the static detonator safety test cable to pins 10(+) and 8(-).

i. Initiate the detonators using a Dupont Blasting Machine Condensor Discharge CD-12-1 (or equal equipment with a 100-μF capacitor charged to 130 Vdc).

j. Remove the fuze from the firing chamber, and detach the fuze from the test setup. Inspect the housing for evidence of rupture, and the witness tape of b above for evidence of perforation. Then, disassemble the fuze and inspect the interior for conformance to 3.4.7.1.

C11

4.7.7.2 Explosive-train output. —

WARNING

The explosive-train output test involves hazardous operations and requires the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 6.4).

The explosive-train output test consists of initiating the fuze on a steel dent block (see Figure 4) and measuring the depth of the dents to determine the output.

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4.7.7.2.1 Instantaneous mode. To determine conformance to 3.4.7.2, modify and test the armed fuze as follows:

WARNING

Proper safety precautions must be observed when making the following modifications and performing the following test (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials.

a. Arm the fuze using the test set.

b. Before handling the fuze, discharge the firing capacitor by shorting pins 5 and 8 together for 2 minutes minimum. Verify that the firing capacitor has been discharged by measuring the voltage across pins 5 and 8; if less than 1 Vdc, the fuze is safe to handle.

c. Verify that the fuze is in the armed condition (see 6.3.2).

d. Place the explosive-train drop fixture (see Drawing SA497753) into a firing chamber capable of containing 2.2 grams of CH-6 (see MIL-C-21723) explosive output.

e. Place the dent block (Figure 4) in a booster cup. Booster cup shall be in accordance with Drawing 2049577 (or equal). Screw the booster cup to the output end of the armed fuze.

f. Install the armed fuze in the drop fixture.

g. Attach a cable to the fuze that will allow the following:

(1) Application of 28 ±1 Vdc to pins 8(-) and 10(+) to charge the firing capacitor.

(2) Connection at pins 4 and 8 to enable the instantaneous detonators.

h. Shorts pins 4 and 8 together. Apply 28 ±1 Vdc to charge the firing capacitor.

i. After charging 30 seconds minimum, pull the fuze release drop wire on the test fixture. (Upon impact, the fuze firing circuit should close, initiating the instantaneous and delay detonators.)

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j. Measure the continuity between pins 8(-) and 12(+) and 8(-) and 9(+) to determine if the fuze is safe to handle (open indication on the 1-kohm scale of the test set).

k. Remove all foreign material from the dent block. (There should be two dents in the dent block).

l. Place the gage probe of a dial-indicator height gage in the deepest part of the first dent and zero the gage indicator.

m. Remove the probe from the dent and take gage readings at two points, each approximately 0.125-inch from the opposite edge of the dent block as shown on Figure 4. Record the average of the two readings.

n. Repeat l and m for the second dent. The average depth of each dent shall conform to 3.4.7.2.

M113

4.7.7.2.2 Inhibit mode. To determine conformance to 3.4.7.2, modify and test the armed fuze as follows:

WARNING

Proper safety precautions must be observed when making the following modifications and performing the following test (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials.

a. Repeat 4.7.7.2.1a, b, and c.

b. Place the explosive-train drop fixture (see Drawing SA 497753) into a firing chamber capable of containing 2.2 grams of CH-6 (see MIL-C-21723) explosive output.

c. Place the dent block (Figure 4) in a booster cup. Booster cup shall be in accordance with Drawing 2049577 (or equal). Screw the booster cup to the output end of the armed fuze.

d. Install the armed fuze in the drop fixture.

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e. Attach a cable to the fuze that will connect pins 5 and 8 and allow application of 28 ±1 Vdc between pins 8(-) and 10(+). Close the firing chamber. Apply 28 ±1 Vdc to pins 8 and 10 charging the firing capacitor.

f. After charging 30 seconds minimum, pull the fuze release drop wire on the test fixture. (Upon impact, the fuze firing circuit should not initiate the detonators.)

g. Before handling the fuze, discharge the firing capacitor by turning off the 28 Vdc and allowing the fuze to set for 2 minutes minimum. Verify that the firing capacitor has been discharged by measuring between pins 5 and 8; if less than 1 Vdc, the fuze is safe to handle.

M114

4.7.7.2.3 Delay mode. To determine conformance to 3.4.7.2, modify and test the armed fuze as follows:

WARNING

Proper safety precautions must be observed when making the following modifications and performing the following test (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials.

a. Repeat 4.7.7.2.1a, b, and c.

b. Place the explosive-train drop fixture (see Drawing SA497753) into a firing chamber capable of containing 2.2 grams of CH-6 (see MIL-C-21723) explosive output.

c. Place the dent block (Figure 4) in a booster cup. Booster cup shall be in accordance with Drawing 2049577 (or equal). Screw the booster cup to the output end of the armed fuze.

d. Install the armed fuze in the drop fixture.

e. Attach a power cable to pins 8(-) and 10(+). Close the firing chamber. Apply 28 ±1 Vdc to charge the firing capacitor.

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f. After charging 30 seconds minimum, pull the fuze release drop wire on the test fixture. (Upon impact, the fuze firing circuit should close, initiating the delay detonators.)

g. Repeat 4.7.7.2.1 j through n.

M115

4.7.8 Visual inspection. Visually inspect the fuze to determine conformance to 3.5 and the general requirements of Section 3.

M116

4.7.9 Drawing inspections. To determine conformance to 3.3.1, perform drawing inspections on parts, subassemblies, and assemblies of the fuze during manufacture, and on the completely assembled, sealed, and pressurized fuze. The sampling plans and acceptance criteria shall be as specified on the drawings.

M117

4.7.10 Packaging inspection. Prior to shipment, inspect the fuze to determine conformance to the requirements of Section 5.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C, and as specified on the drawings listed on Drawing 642AS5639.

5.1.1 Level C. Fuzes shall be packaged in accordance with the drawings listed on Drawing 642AS5639 and in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and as specified on the drawings listed on Drawing 642AS5639.

5.2.1 Level C. Fuzes, packaged in accordance with 5.1.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing and shipping containers shall conform to 49 CFR 171-178.

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5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with the drawings listed on Drawing 642AS5639 and MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The fuze is intended for use as a major component of the Harpoon Missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification.

b. Government-furnished property. The contracting officer should arrange to furnish the property cited in 3.1.1.1 and 3.1.1.2.

c. Government-loaned property. The contracting officer should arrange to loan the property cited in 3.1.2.

d. First article. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 30 fuzes as specified in 4.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

e. Responsibility for inspection, if other than as specified in 4.1.

f. Inspection facility, if other than as specified in 4.1.

g. Independent facility to perform the inspections of Figure 9 (see 4.4, 4.5.2, and 4.7.5).

h. Inspection lot size, if other than as specified in 4.5.1.

i. Assignment of serial numbers, if other than as specified in 4.5.1.1.

j. Government final inspection gage drawings (see 4.1.1.4).

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k. Disposition of armed fuzes which repeatedly fail to reset to safe position, if other than as specified in 4.3.1.

l. Disposal of fuzes subjected to inspections of Figure 9 (see 4.7.5).

m. Special marking (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below should be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below should be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
4.1.1.1	Calibration Plan	DI-R-7064
4.3.2	Test Plan and Test Procedures	DI-T-5204
4.3.3	Inspection Lot Report	UDI-R-20344B
4.3.4	Production Summary Report	DI-A-5030B
4.3.5	Inspection Record	UDI-M-20404

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Safe condition and safe position. Visually, the fuze is considered to be in the safe condition or safe position when a white letter "S" on a green background is centered in the viewing window located in the housing. Electrically, the fuze is considered to be in the safe condition or safe position when no explosive components of the fuze initiate during the test of 4.7.4.

6.3.2 Armed condition and armed position. Visually, the fuze is considered to be in the armed condition or armed position when a white letter "A" on a red background is centered in the viewing window located in the housing. Electrically, the fuze is considered to be in the armed condition or armed position when, as measured on the test set, the limits of Tables IV and V are met under the conditions specified in the tables.

6.3.3 Safe. Safe is defined as the ability of the fuze to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service

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life which will nullify the required safety conditions of service use. For nonservice use, safe is defined as the ability of the fuze to maintain its safety features in a condition which will permit disposal without injury to personnel.

6.3.4 Operable. Operable is defined as the ability of the fuze, with specified inputs, to perform to completion its specified function and sequence, and produce the specified outputs within the required operational period or at the specified point of time.

6.3.5 Instantaneous mode. The instantaneous mode is defined as the instantaneous initiation of the MARK 71 MOD 1 Detonators when a target is sensed. The instantaneous mode was designed to provide flexibility for future fuzing requirements.

6.3.6 Inhibit mode. The inhibit mode prevents the initiation of the detonators due to impact.

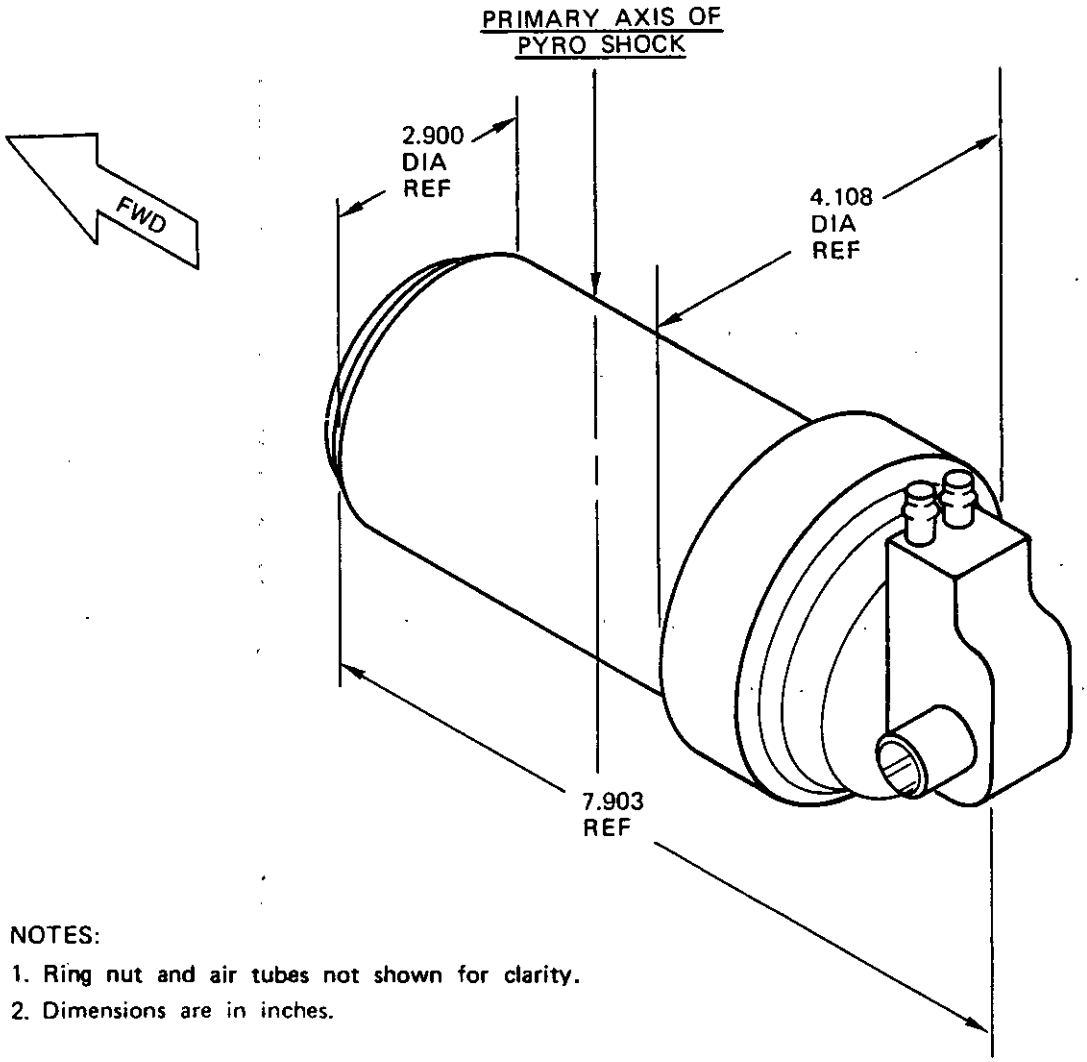
6.3.7 Delay mode. The delay mode is defined as the initiation of the MARK 85 MOD 0 Detonators after a specified time delay following impact.

6.3.8 Alternate mode. The alternate mode is defined as the initiation of the MARK 85 MOD 0 Detonators and the MARK 71 MOD 1 Detonators (if selected) when 28 ±1 Vdc is applied to pin 1 of the fuze through a 22-kohm resistor.

6.4 Explosives safety precautions. Minimum explosives safety precautions for use by the contractor are detailed in DoD Instruction 4145.26M, DoD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material.

Preparing activity
Navy-AS
(Project 1336-N430)

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NOTES:

1. Ring nut and air tubes not shown for clarity.
2. Dimensions are in inches.

FIGURE 1. FMU-141/B guided missile fuze general configuration.

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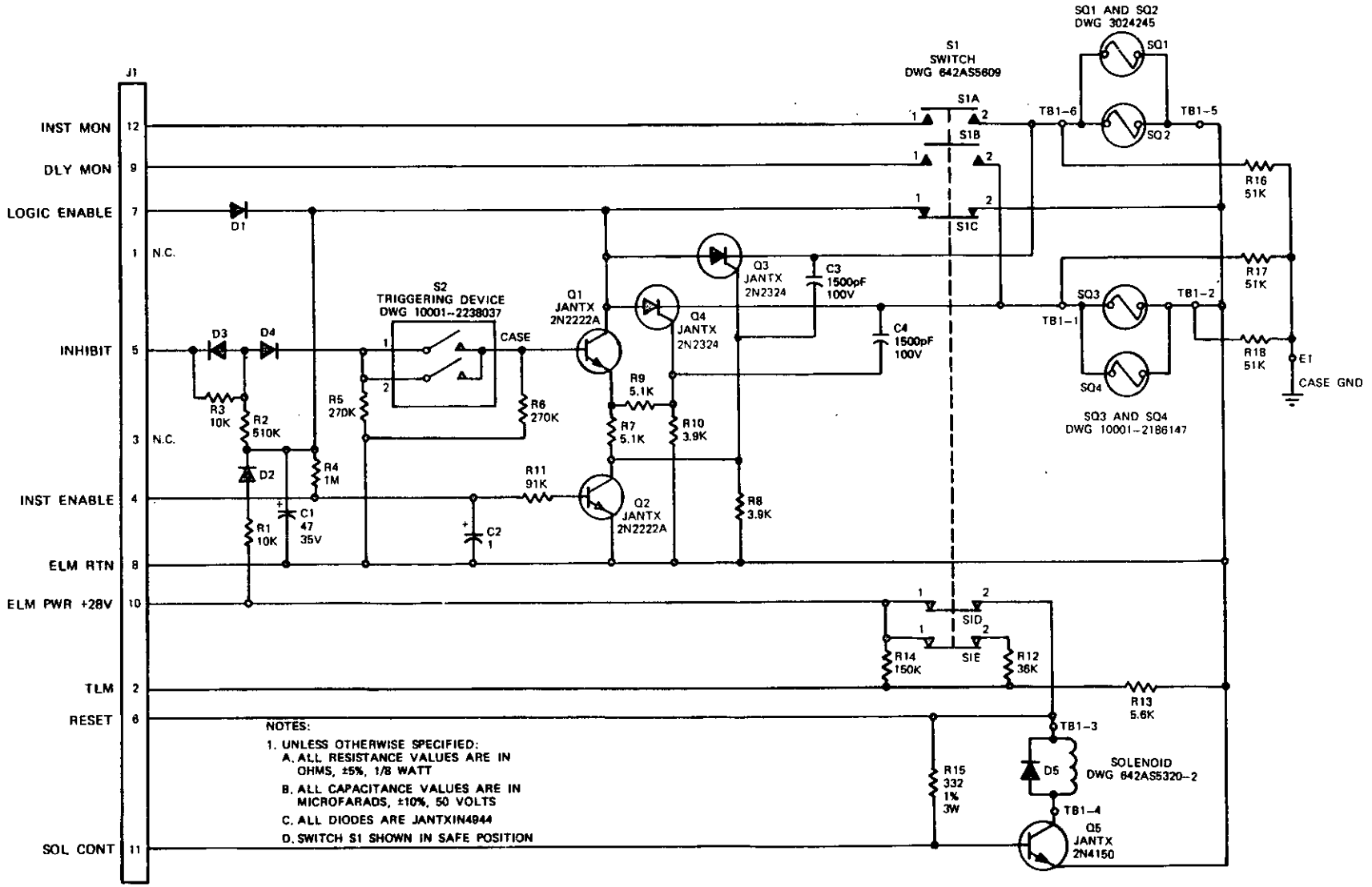


FIGURE 2. Electrical schematic (safe condition).

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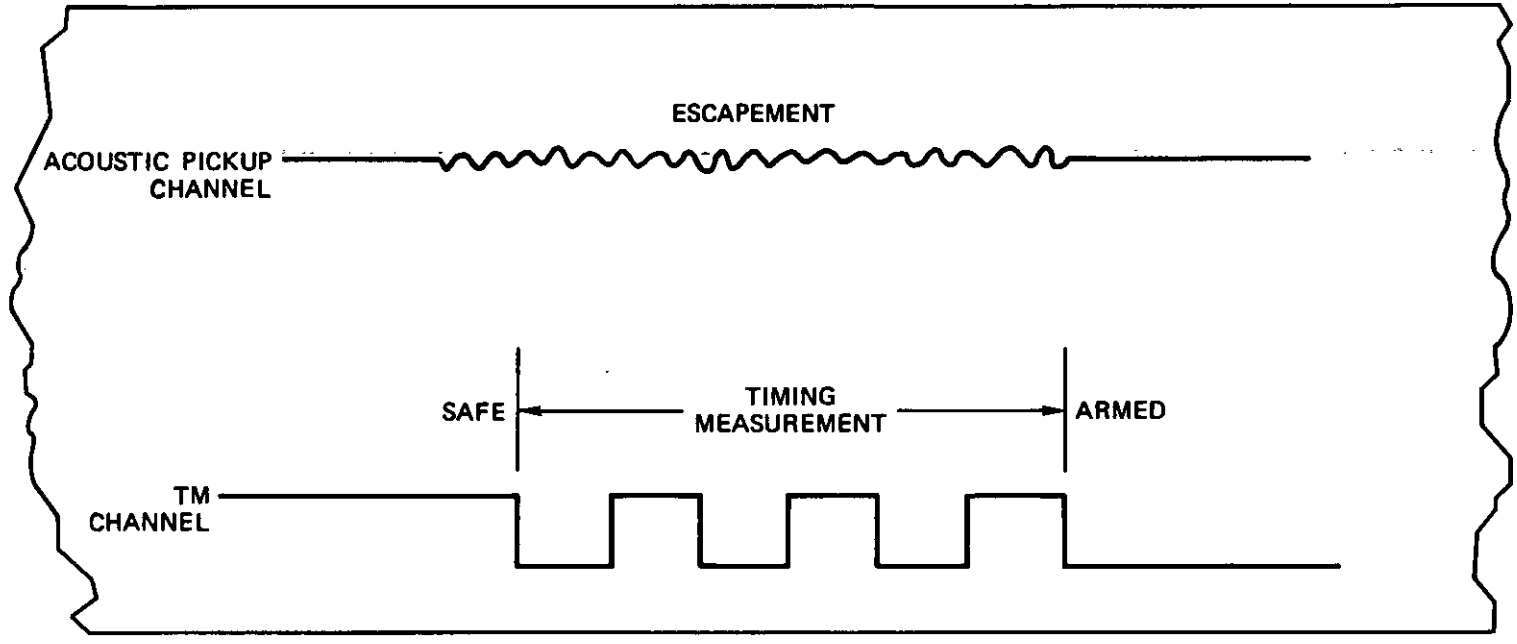
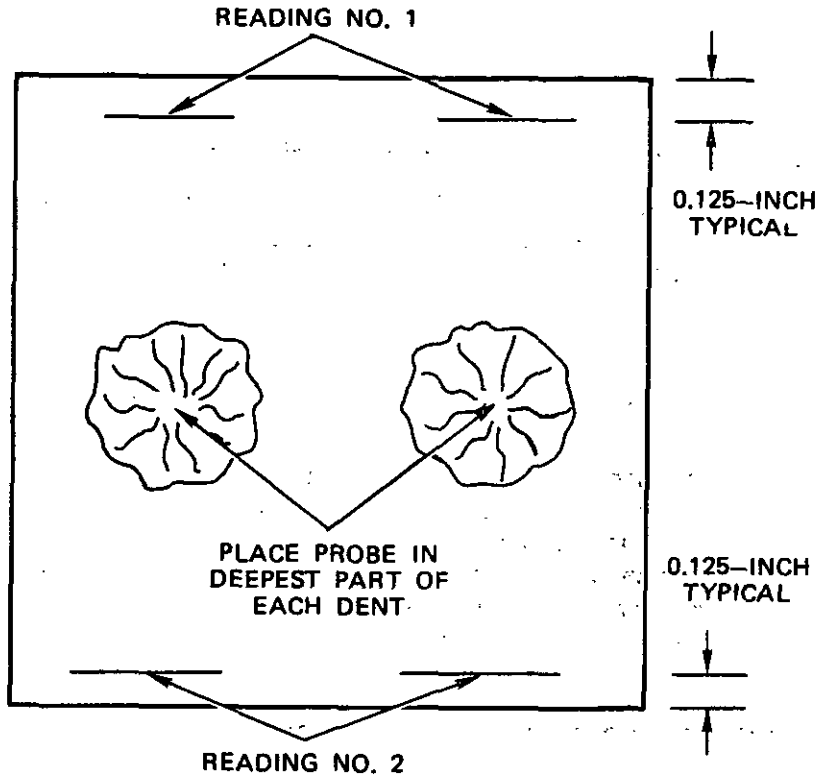


FIGURE 3. Sample of test set fuze timing output record chart.

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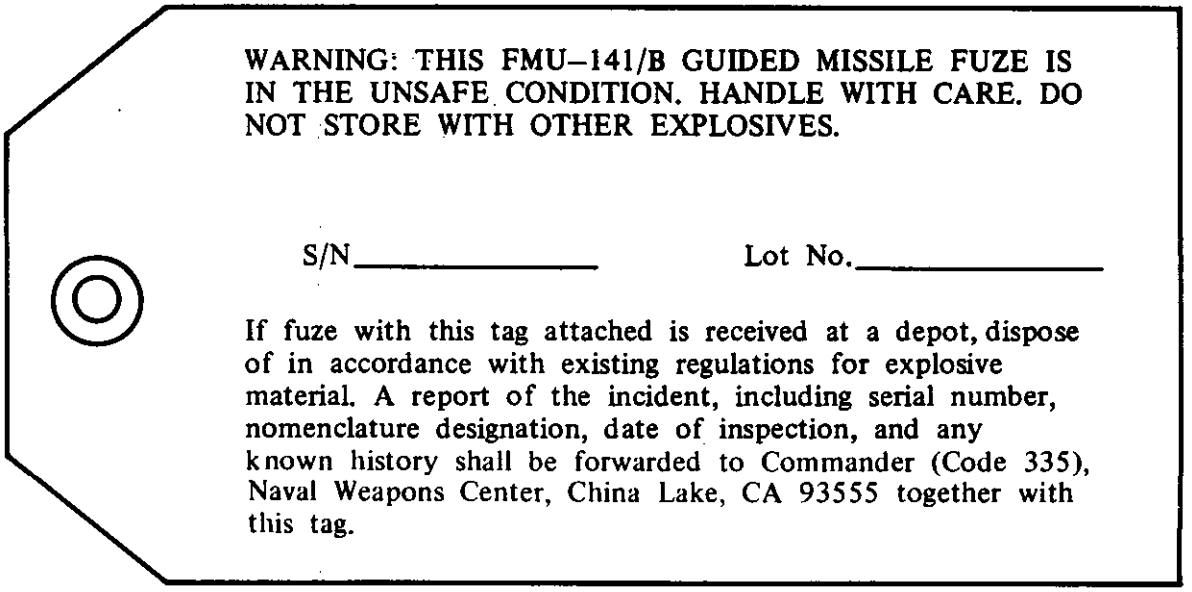


NOTES:

1. Dent block shall be steel, cold-finished bar in accordance with ASTM–A 108, Grade 1018 or 1020, Rockwell hardness B70 to B95, surface condition as rolled. Size, 1.750 ±0.125 inches square, 1.250 ±0.125 inches thick. Block shall be clean and free from rust and burrs.
2. Dent block shall be placed inside a booster cup. Booster cup shall be in accordance with Drawing 2049577 (or equivalent) for the test of 4.7.7.2.2 and 4.7.7.2.3.

FIGURE 4. Dent block and dent measurement.

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NOTE: Tag shall be in accordance with UU-T-81, Type B, Grade 15SU, Size 7 or 8, Color red. The required data may be printed or stamped thereon.

FIGURE 5. Unsafe condition tag.

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INSPECTION RECORD
FMU-141/B GUIDED MISSILE FUZE

CONTRACT NO. _____ MANUFACTURER _____

PREPRODUCTION SAMPLE ☐
INSPECTION LOT ☐
REWORK ☐

S/N _____ LOT NO. _____

INSPECTOR _____ DATE _____

ELECTRICAL RESISTANCE

SAFE CONDITION

TEST	CONFIG A	CONFIG B	CONFIG C	TEST	CONFIG A
S1				S8	
S2				S9	
S3				S10	
S4				S11	
S5				S12	
S6				S13	
S7					

ARMED CONDITION

CONFIG B	CONFIG C	TEST	CONFIG A	CONFIG B
[redacted]		A1		
		A2		
		A3		
		A4		
		A5		

CHECKED BY _____
DATE _____

DRAWING INSPECTIONS

DWG 642AS5422 VERIFY NOTE 5
CHECKED BY _____ DATE _____

DWG 642AS5610 VERIFY NOTE 12
CHECKED BY _____ DATE _____

DWG 642AS5618 VERIFY NOTE 7

TEST	OHMS	TEST	OHMS
8		16	
9		18	
15			

CHECKED BY _____ DATE _____

PARTS/SUBASSEMBLIES
VERIFIED BY _____ DATE _____

DWG 642AS5610 – FINAL ASSEMBLY

VERIFIED BY _____ DATE _____

MECHANICAL OPERABILITY

TEST	RESULT	TEST	RESULT
ALL GO		TIMING	
NO GO		BURST	

CHECKED BY _____ DATE _____

SEAL

LEAK RATE _____ STD ATM $CM^{3/S}$
CHECKED BY _____ DATE _____

SAFETY DETERMINATION
VERIFIED BY _____ DATE _____

FINAL VISUAL INSPECTION

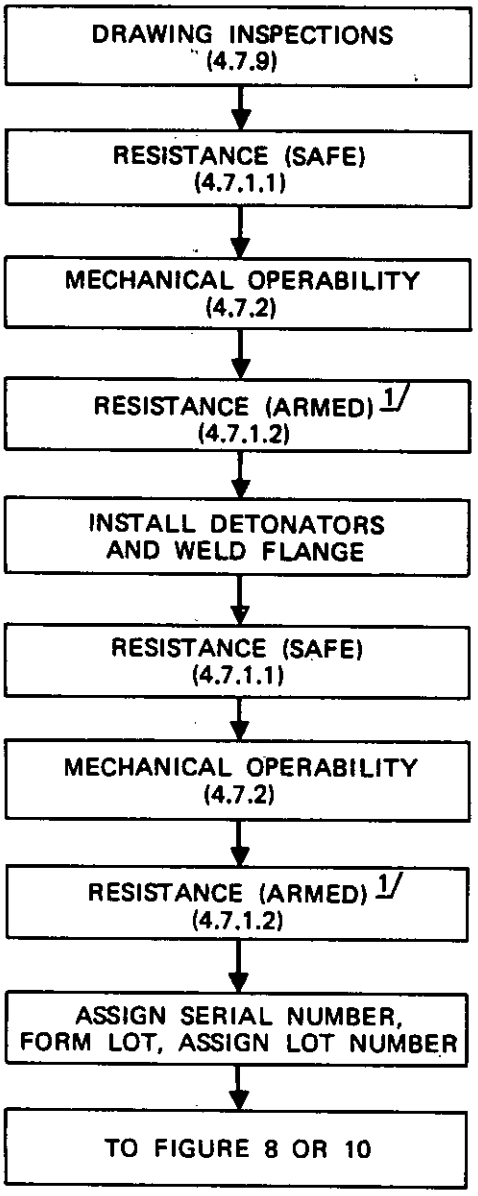
		YES	NO
(1)	EXTERNAL DAMAGE	☐	☐
(2)	FUZE IN SAFE CONDITION (WHITE "S" VISIBLE THROUGH WINDOW)	☐	☐
(3)	SAFETY INSPECTION TAG ATTACHED BY LEAD AND WIRE SEAL, PROPERLY FILLED IN, AND ALL SIGNATURES PRESENT	☐	☐

GOVERNMENT REPRESENTATIVE:

_____ _____
SIGNATURE DATE

FIGURE 6. Sample inspection record.

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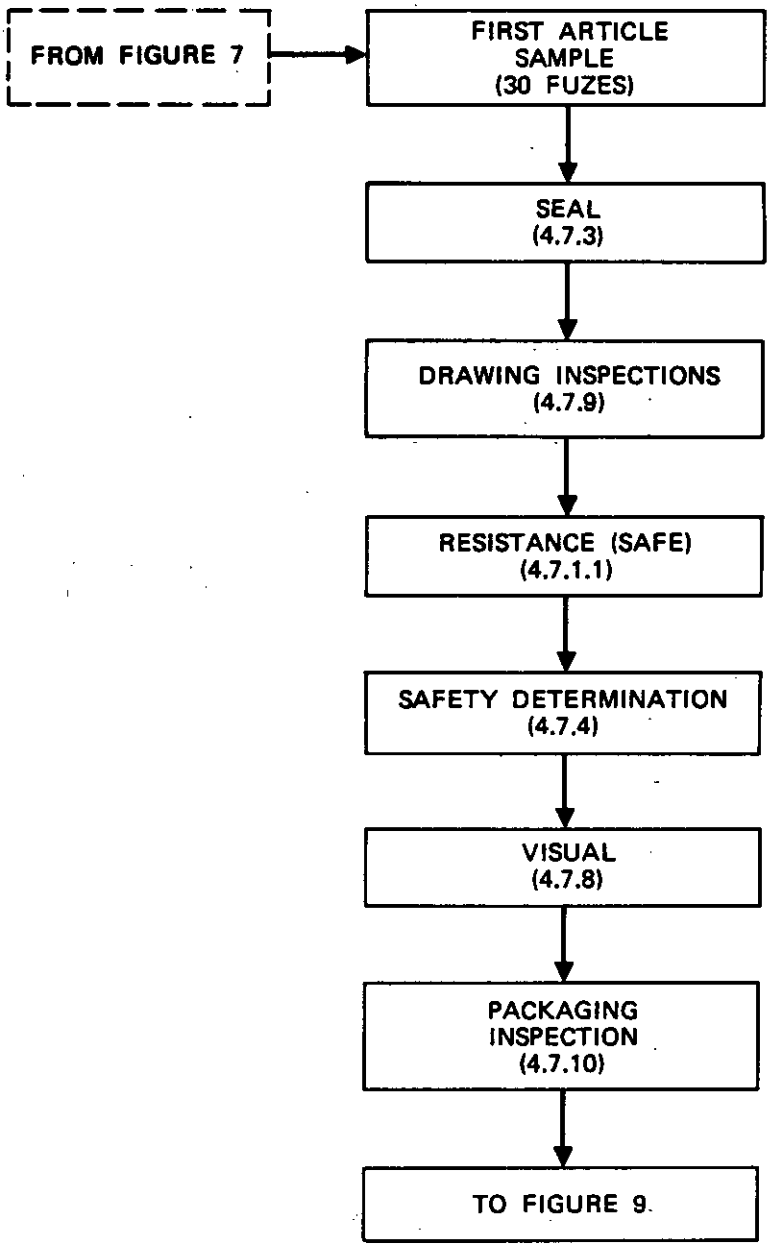
NOTES:

1. Test results shall be recorded on the inspection record of Figure 6.

1/ Reset the fuze to the safe condition after test (see 4.3.1).

FIGURE 7. Individual unit acceptance inspection sequence.

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NOTE: All test data shall be recorded on the inspection record (see Figure 6).

FIGURE 8. First article sample inspection sequence.

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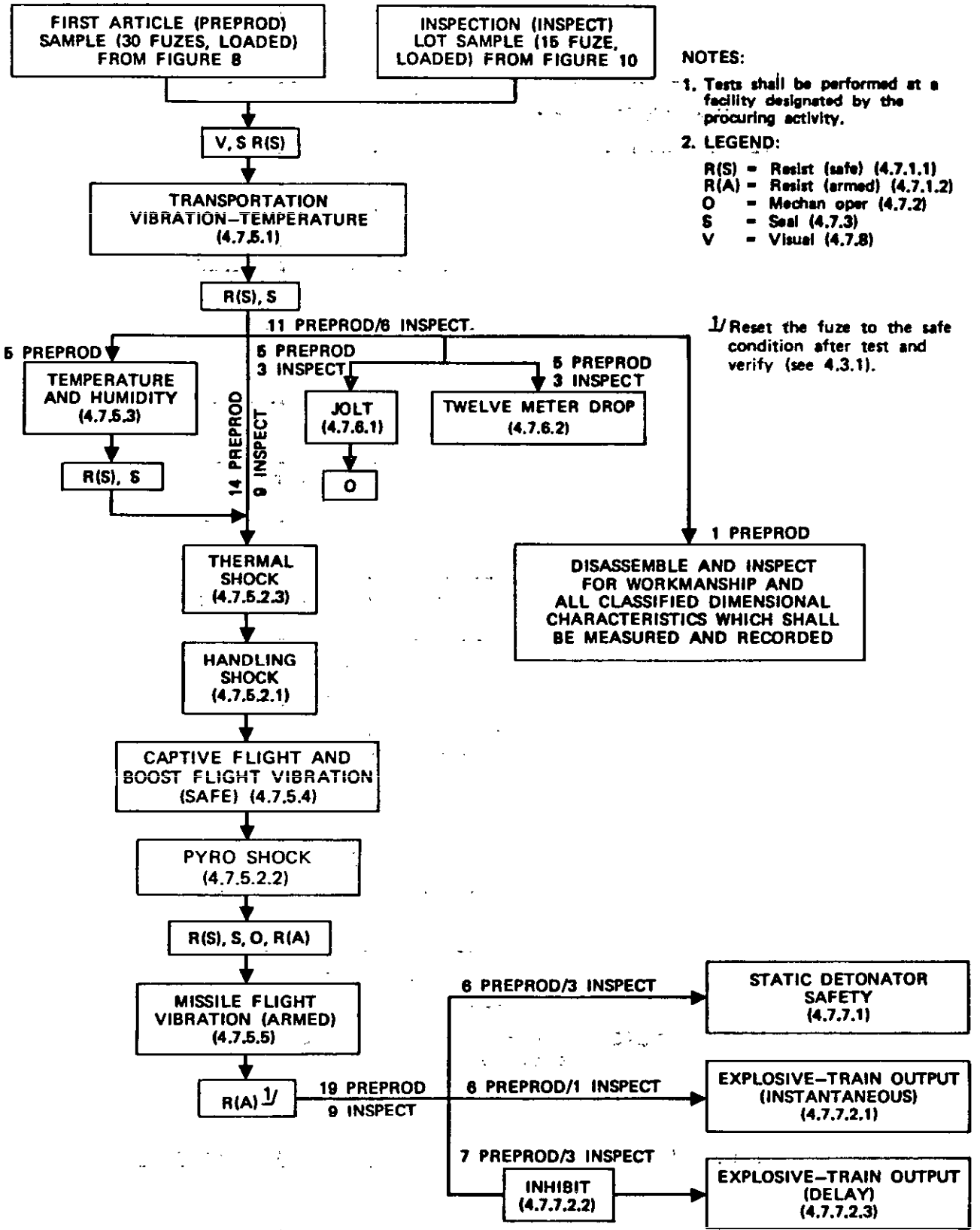
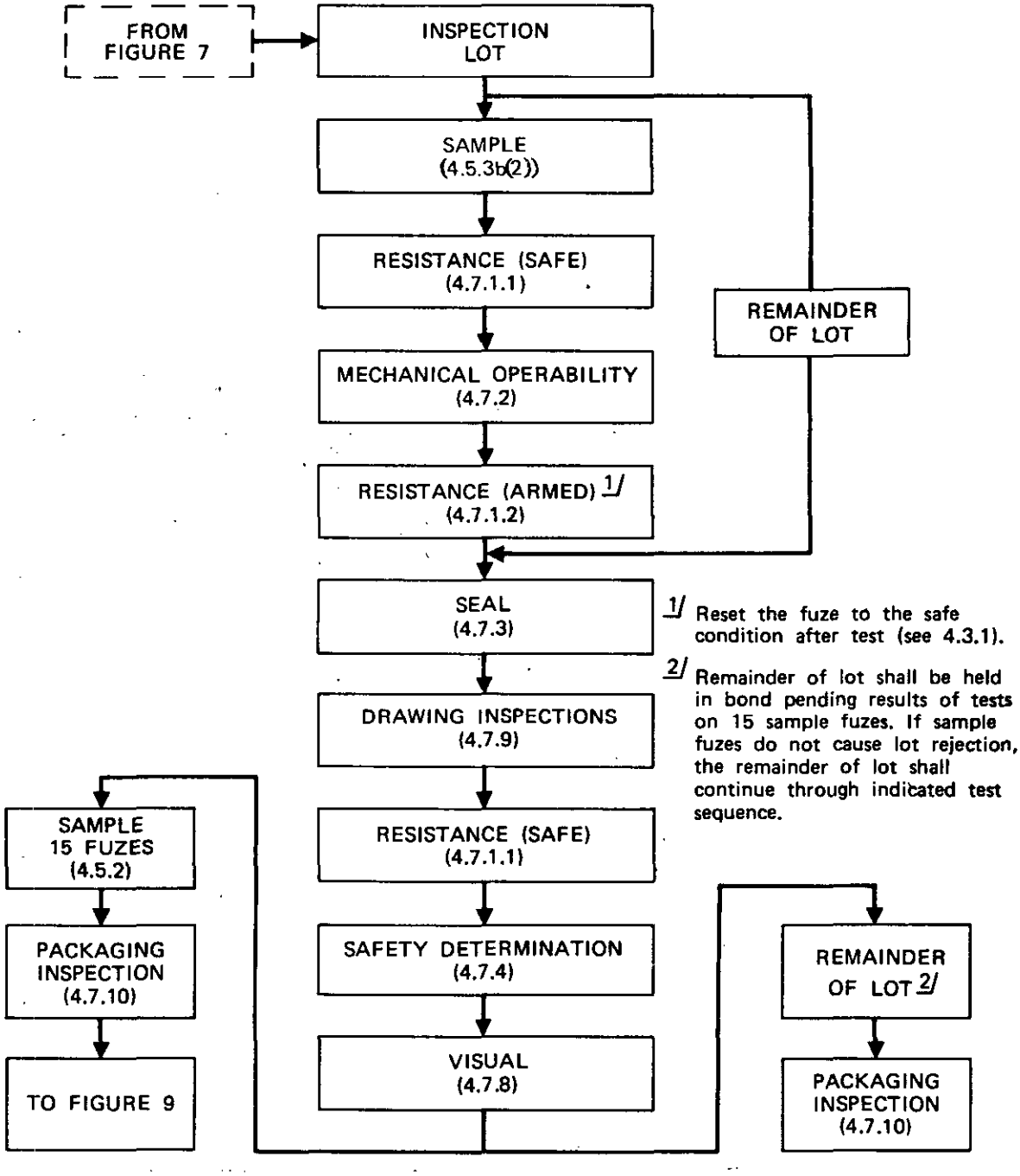


FIGURE 9. Environmental and explosive-train inspection sequence.

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NOTE: Test data shall be recorded on inspection record (see Figure 6).

FIGURE 10. Inspection lot inspection sequence.

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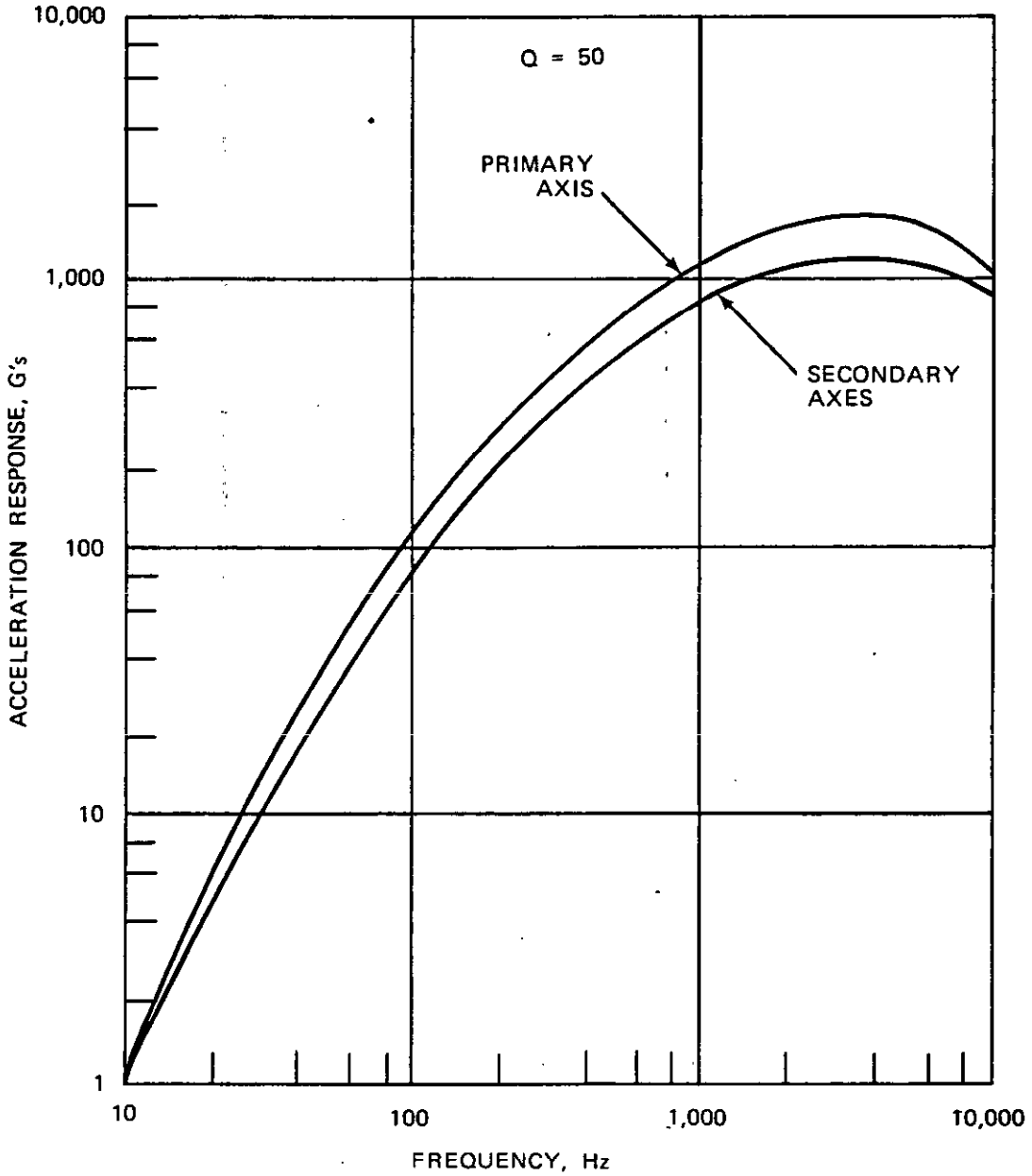


FIGURE 11. Primary and secondary pyro shock spectrum.

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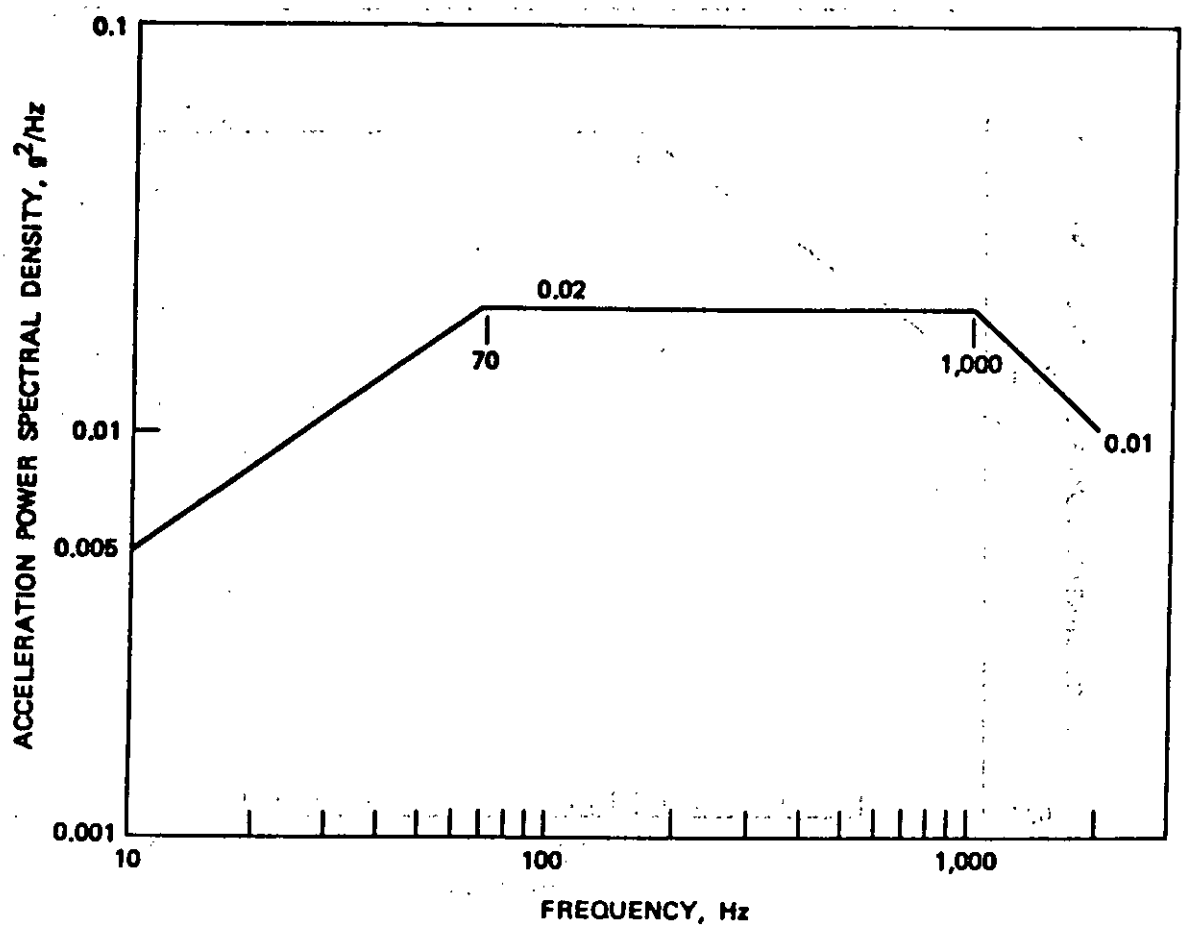


FIGURE 12. Captive and missile flight vibration.

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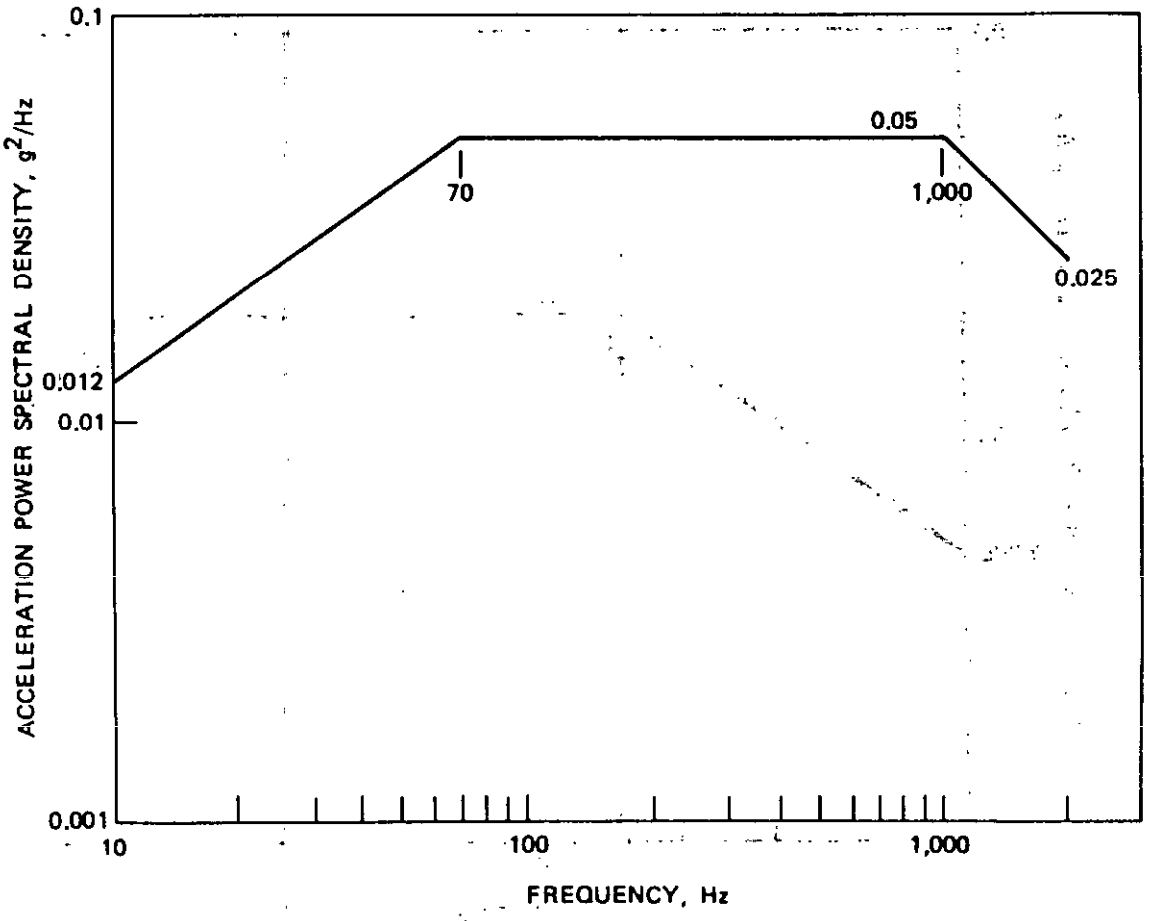
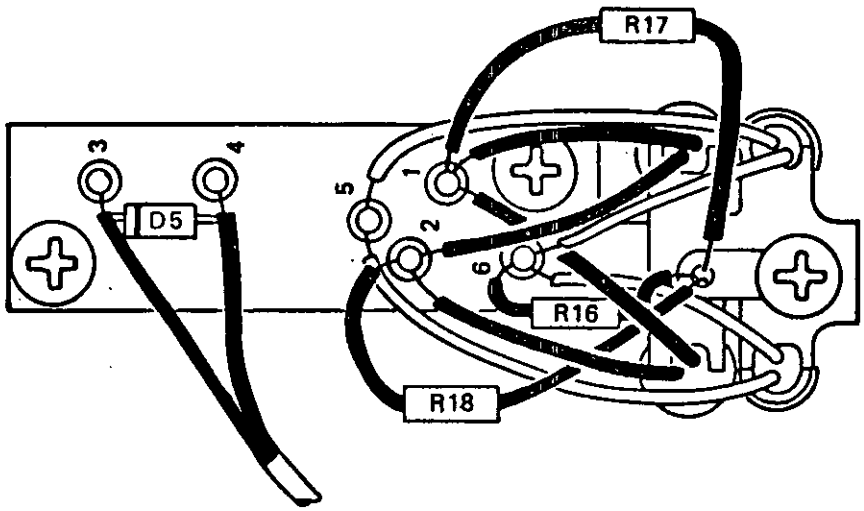


FIGURE 13. Boost flight vibration.

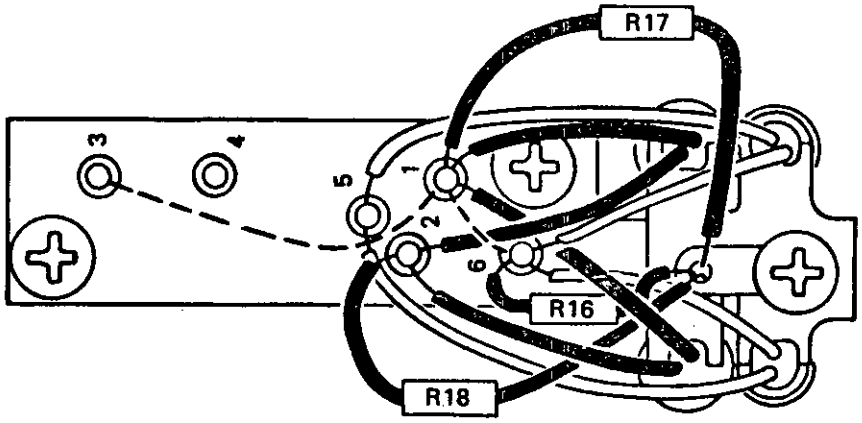
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REGULAR WIRING



(SEE DRAWING 642AS5610)

MODIFIED WIRING



— — — ADDED JUMPER WIRES

FIGURE 14. Terminal board TB1 modification for static detonator safety test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-85475(AS)	Fuze, Guided Missile, FMU-141/B

3a. NAME OF SUBMITTING ORGANIZATION

Naval Weapons Center
China Lake, CA 93555

b. ADDRESS *(Street, City, State, ZIP Code)*

China Lake, CA 93555

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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